

References

1. Mantri, S.; et al. Proceedings of the World Summit on Parkinson's Disease. *npj Parkinson's Dis.* **2025**, *11*, 293. <https://doi.org/10.1038/s41531-025-01123-8>.
2. Stocchi, F.; Bravi, D.; Emmi, A.; Antonini, A. Parkinson Disease Therapy: Current Strategies and Future Research Priorities. *Nat. Rev. Neurol.* **2024**, *20*, 695–707. <https://doi.org/10.1038/s41582-024-01034-x>.
3. Kim, T. Y.; Lee, B. D. Current Therapeutic Strategies in Parkinson's Disease: Future Perspectives. *Mol. Cells* **2025**, *48*, 100274. <https://doi.org/10.1016/j.mocell.2025.100274>.
4. Tolosa, E.; Vila, M.; Klein, C.; Rascol, O. LRRK2 in Parkinson Disease: Challenges of Clinical Trials. *Nat. Rev. Neurol.* **2020**, *16*, 97–107. <https://doi.org/10.1038/s41582-019-0301-2>.
5. Zhu, H.; Hixson, P.; Ma, W.; Sun, J. Pharmacology of LRRK2 with Type I and II Kinase Inhibitors Revealed by Cryo-EM. *Cell Discov.* **2024**, *10*, 10. <https://doi.org/10.1038/s41421-023-00639-8>.
6. Liessmann, F.; Eisenhuth, P.; Füll, A.; Vu, O.; Moretti, R.; Meiler, J. Cache: Utilizing Ultra-Large Library Screening in Rosetta to Identify Novel Binders of the WD-Repeat Domain of Leucine-Rich Repeat Kinase 2. *J. Cheminform.* **2025**, *17*, 145. <https://doi.org/10.1186/s13321-025-01084-3>.
7. Taylor, S. S.; et al. Kinase Domain Is a Dynamic Hub for Driving LRRK2 Allosterity. *Front. Mol. Neurosci.* **2020**, *13*, 538219. <https://doi.org/10.3389/fnmol.2020.538219>.
8. Li, F.; et al. CACHE Challenge #1: Targeting the WDR Domain of LRRK2, a Parkinson's Disease-Associated Protein. *J. Chem. Inf. Model.* **2024**, *64*, 8521–8536. <https://doi.org/10.1021/acs.jcim.4c01267>.
9. Mettu, A.; et al. Functionally Active Modulators Targeting the LRRK2 WD40 Repeat Domain Identified by FRASE-Bot in CACHE Challenge #1. *Chem. Sci.* **2025**, *16*, 3430–3439. <https://doi.org/10.1039/D4SC07532C>.
10. MolPort. *Chemical Compound Database and Marketplace*. <https://www.molport.com/> (accessed April 2026).
11. Enamine Ltd. *EnamineStore: Chemical Compound Database*. <https://enaminestore.com/> (accessed April 2026).
12. Meng, X.-Y.; Zhang, H.-X.; Mezei, M.; Cui, M. Molecular Docking: A Powerful Approach for Structure-Based Drug Discovery. *Curr. Comput.-Aided Drug Des.* **2011**, *7*, 146–157. <https://doi.org/10.2174/157340911795677602>.
13. Rocklin, G. J.; Mobley, D. L.; Dill, K. A.; Hünenberger, P. H. Calculating the Binding Free Energies of Charged Species Based on Explicit-Solvent Simulations Employing Lattice-Sum Methods. *J. Chem. Phys.* **2013**, *139*, 184103. <https://doi.org/10.1063/1.4826261>.
14. Jerabek-Willemsen, M.; et al. MicroScale Thermophoresis: Interaction Analysis and Beyond. *J. Mol. Struct.* **2014**, *1077*, 101–113. <https://doi.org/10.1016/j.molstruc.2014.03.009>.
15. Drescher, D. G.; Selvakumar, D.; Drescher, M. J. Analysis of Protein Interactions by Surface Plasmon Resonance. In *Advances in Protein Chemistry and Structural Biology*; Donev, R., Ed.; Academic Press, 2018; Vol. 110, pp 1–30. <https://doi.org/10.1016/bs.apcsb.2017.07.003>.

Code & Project Repository

GitHub Repository: https://github.com/nancyihechu/LRRK2_WD40_Analog_Scoring

Implementation of weighted and SKNN-based scoring for analog prioritization in LRRK2 WD40 drug discovery.

Contact & Networking

LinkedIn: <https://www.linkedin.com/in/nancy-ihechu/>

I look forward to discussion, collaboration, or feedback

Nancy Ihechu

Email:nancyihechu@gmail.com || Phone: +1 573 814 8366|| [LinkedIn](#) || [GitHub](#)

SUMMARY

Computational chemist with experience developing machine learning methods for drug discovery. Skilled in free energy calculations, and molecular modeling. Passionate about developing computational methods that accelerate scientific discovery.

EDUCATION

Ph.D. Chemistry (Computational Drug Discovery)
University of Missouri, USA

Jan.2026 – Present

M.Sc. Chemistry/Chemo-Informatics
University of Strasbourg, France (AI in Chemistry)
University of Ljubljana, Slovenia (Chemistry)

Sep.2023 – July.2025

B.Sc. Chemistry
Imo State University, Nigeria

Jan.2015 – Feb.2019

WORK EXPERIENCE

Graduate Research Assistant
University of Missouri-Columbia, USA

May 2025 – Present

- Secured a competitive \$5,000 research grant to develop a quantum machine learning model for classifying and prioritizing potential drug candidates for LRRK2 PROTAC
- Developed and benchmarked ensemble machine learning models using a DNA-encoded library (DEL) of over 7.7 million compounds to identify experimentally confirmed PGK2 actives for a male contraceptive drug discovery project
- Evaluated the binding thermodynamics of over 100 ligand–protein complexes using Absolute Binding Free Energy calculations in GROMACS, supporting hit candidate prioritization for Parkinson's disease drug discovery
- Applied similarity-based virtual screening to the Enamine REAL library of over 100 billion compounds, identifying LRRK2 candidate molecules for downstream experimental validation
- Reduced molecular dynamics simulation time by 60% through GPU-optimized HPC pipelines, Slurm job scheduling, and large-scale parallel execution

Summer Intern
Novartis Slovenia

March.2025 – April.2025

- Developed automated Python workflows for Host Cell Protein (HCP) data extraction, transformation, visualization, and reporting, contributing to a project aimed at reducing biological data processing time
- Improved accessibility of HCP datasets by developing interactive dashboards for real-time monitoring of protein data and project progress, supporting cross-functional research teams
- Communicated analytical insights by presenting clear visualization of data-driven findings that supported project planning and operational efficiency

Laboratory Analyst
Standards Organization of Nigeria

Dec.2019 – July.2023

- Performed chromatography and mass spectrometry analyses on environmental samples, supporting analytical data generation, quality assessment, and regulatory testing
- Improved interpretation of analytical results by developing Python- and Power BI-based workflows that automated quality-control reporting and supported a team of 10 laboratory scientists
- Maintained regulatory compliance and scientific documentation by preparing quality-control records, interpreting analytical results, and ensuring accurate laboratory record-keeping
- Managed laboratory inventories, coordinating equipment maintenance schedules, and ensuring the availability of research supplies

PUBLICATION

PRESENTATIONS

American Chemical Society Regional Meeting

Oct. 2025

- Title: Anti-Parkinson's Therapeutic Discovery: A Computational Approach
- Description: Poster Presentation on prioritization workflow in the early phase drug discovery

HONORS AND AWARDS

Erasmus Mundus Double Master's Degree Scholarship

Sep. 2023

- Description: Competitive European Union-funded scholarship awarded for a joint international master's program in Chemistry and Chemoinformatics.

AFFILIATION

- American Chemical Society (ACS) – Member
- Machine Learning Innovation Network For Research to Advance MEDicinal Chemistry (MAINFRAME) – Member